Ohio University: Department of Physics and Astronomy
Physics 6021, 2024, Homework III, *Due 15 April, 2024*

1. Zett 8.1
2. Zett 8.4
3. Zett 8.12
4. Consider three particles of mass *m* in a 1-D square well of side a with impenetrable walls. We have discussed the solution to the single-particle problem several times. Assume that the particles interact with each other according to:

$$V = V_0 b[\delta(x_1 - x_2) + \delta(x_2 - x_3) + \delta(x_1 - x_3)]$$

Where *b* has units of length. Use first order perturbation theory to calculate the energy levels of the system assuming:

a) Particles are not identical
b) Identical particles of spin zero, with at least two particles in the same single-particle state
c) Identical particles of spin ½ in a state of total spin 3/2

5. *Thomas-Fermi model of an atom.* In the Thomas-Fermi approximation, the kinetic energy is expressed as a functional of the electron density $\rho(x)$, and takes the form: $T[\rho] = c_1 \int dV\, \rho(x)^{5/3}$. Here, c_1 is a constant, the integral is three-dimensional and is assumed to extend over the charge distribution $\varrho(x)$. Consider a neutral atom with nucleus of positive charge *Z=Ne*, fixed at the origin.
   a. Write down the energy E as a functional of $\rho(x)$. Include contributions of the e-e repulsion and the attraction to the nucleus. *Get your signs right.*
   b. Introduce a Lagrange multiplier λ to enforce the constraint condition that $\int dx\, \rho(x) = N$, where *N* is the number of electrons. Use the method of Lagrange multipliers and the Euler-Lagrange equation to minimize E[ρ] subject to the constraint. *The result is a complicated equation involving ρ and λ: don't try to solve it.*